Marks scored:

KATHMANDU UNIVERSITY
End Semester Examination [C]
November, 2018

Level :B. Pharm.
Year : II

Course : CHEM 203
Semester: II

Exam Roll No. :

Time: 30 mins.

F. M. : 20

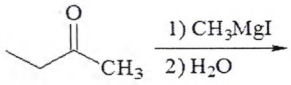
Registration No.:

Date : NOV 15 2018

SECTION "A"
[20 Q.×1=20 marks]

Mark X on the most appropriate answer/s.

1. The following transformation using a Grignard reagent will yield

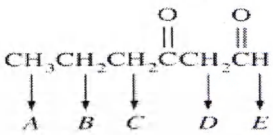


[ ] [ ] [ ] [ ]

2. Which amino acid has the lowest isoelectric point?

[ ] glycine [ ] serine [ ] glutamic acid [ ] lysine

3. The most acidic hydrogen in the given structure is



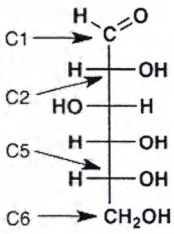
[ ] A [ ] B [ ] C [ ] D

4. 4-O-(α-D-glucopyranosyl)-D-glucopyranose is commonly known as

[ ] (+) –Maltose [ ] (+) –Lactose [ ] (+) –Celloboise [ ] (+) -Sucrose

5. Which of the following method of amino acid synthesis has the highest yield ?

[ ] Amination of α-halo acid [ ] Modified Malonic ester synthesis
[ ] Gabriel phtahlimide method [ ] Phtahlimidomalonic ester method



D-glucose

6. Which carbon becomes the anomeric carbon of glucose in its pyranose form?

[ ] C1 [ ] C2 [ ] C5 [ ] C6

7. Electropilic substitution of Furan is favorable at.....position/s.

[ ] 2,3 [ ] 2,4 [ ] 2,5 [ ] 2

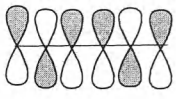
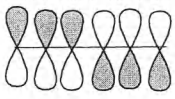
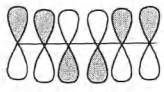
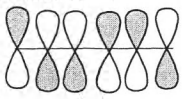
8. Propanone on reaction with Grignard reagent gives

[ ] Primary alcohol [ ] Secondary alcohol
[ ] Tertiary alcohol [ ] Quaternary alcohol

9. Amylopectin has
[ ] β-1-4 and β-1-6 linkage
[ ] α-1-4 and β-1-6 linkage
[ ] β-1-4 and α-1-6 linkage
[ ] α-1-4 and α-1-6 linkage

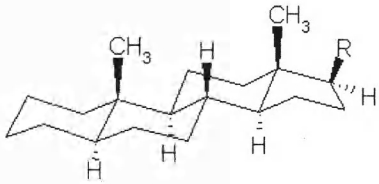
10. Which of the following compounds does not give the aldol reaction?
[ ] Methanal [ ] Ethanal [ ] Propanal [ ] Butanal

11. The orbital of the 1,3,5-hexatriene involved in thermal electrocylic reaction is

[ ]  [ ]  [ ]  [ ] 

12. Which of the following is not true?
[ ] [4+2] thermal cycloaddition takes place on supra, supra mode
[ ] [4+2] photochemical cycloaddition takes place on supra, supra mode
[ ] [2+2] thermal cycloaddition takes place on antra, supra mode
[ ] [4+2] photochemical cycloaddition takes place on antra, antra mode

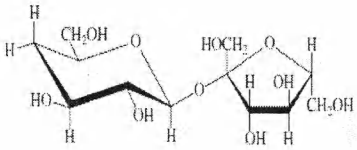
13. The position of $-CH_3$ groups in the given steroid are.....
[ ] 5, 10 [ ] 6,12
[ ] 18,19 [ ] 12, 16



14. Which of the following statement is not true?
[ ] Pyridine is more reactive towards electrophiles than benzene
[ ] Both pyridine and benzene undergo reactions with electrophiles, but pyridine always forms *N*-derivatives
[ ] Pyridine is less reactive towards electrophiles than benzene
[ ] Both pyridine and benzene behave similarly towards electrophiles

Fill in the blanks.

15. Non peptide portion present in the protein molecule is called .....

16. This is the structure  of.....

17. Pyridine on Sulphonation yields.....(write structural formula)

18. In 1,3,5, hexatriene , the excited state HOMO is present at the energy level.....

19. In the enzyme Chymotrypsin the –OH group involved in the enzyme action is associated with the amino acid.....(give the structural formula)

20. [1,5] carbon sigmatropic migration in suprafacial mode occurs with.....of configuration.

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NOV 1 5 2018

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Semester: II
F.M. : 55

SECTION "B"
[55 Marks]

Attempt *ALL* questions.

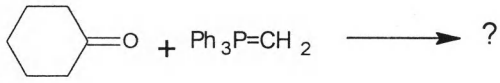
1. Give the mechanism for the following reactions. [4×2.5=10]
   a. Malonic ester synthesis
   b. Claisen condensation
   c. Diels Alder reaction
   d. Reaction of Methyl magnesiumcholride with butan-2-ol followed by hydrolysis.

2. Explain the following statements_ [6×2=12]
   a. Electrophilic substitution at Thiophene is favorable at 2-position than at 3-position.
   b. Pyridine is more basic than Pyrrole but less basic than aliphatic amines.
   c. D-(+)-glucose has six member ring structure.
   d. (+)-Lactose is reducing sugar.
   e. Epimers give the same Osazone.
   f. Ectrocyclic cyclalization of conjugated diene proceeds in conrotatory motion in photochemical condition.

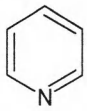
3. Give the chemical reactions involved in the following process. [5×2=10]
   a. Conversion of Aldopentose to Aldohexose
   b. Reaction of Celloboise with Phenyl hydrazine followed by hydrolysis
   c. Reaction of a peptide chain with 2, 4-Dinitroflurorobenzen then acid hydrolyzed.
   d. Reaction of triacylglycerol with Ehanol in basic medium
   e. Preparation of amino acid by Gabriel phthalimide synthesis.

4. Explain the following. [2+2+2+2+3=11]
   a. Cyclodextrin
   b. Mutarotation
   c. Chymotripsin
   d. Structure of Stereiods
   e. Biosynthesis of fatty acid

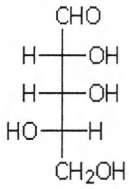
5. Give the product/s of the following reactions. [8 × 1.5=12]

a. 

b. $HOOC\text{-}CH_2CH_2\text{-}COOH + Br_2 \longrightarrow ? + NH_3\,(excess) \longrightarrow$

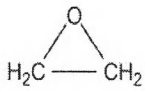
c.  $+ \; NaNH_2 \longrightarrow ?$

d. $C_2H_5Mg\,Br + CH_3CO\,C_2H_5 \longrightarrow ?$

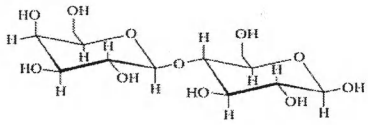
e. 

$C_6H_5NHNH_2$

$\longrightarrow$

f. $CH_3(CH2)_{10}\,CH_2OH +$  base ? $\longrightarrow$

g. $BrCH_2CH_2CH_2CH_2Br + Na_2S \xrightarrow[\text{heat}]{} ?$

h. 

$\xrightarrow[\text{ii. hydrolysis}]{\text{i. } Br_2/H_2O}$